

Executive Summary — Picking a Speech-to-Text Engine for Noisy Support Calls

Assignment 1 · Research-Technical (AI/ML) · Immverse AI Parth Wadekar · parthwadekar40@gmail.com · September 2026

The question. Which ASR model should power a voice AI assistant on customer-support calls? These calls are noisy by nature: busy floors, mixed accents, telephony audio. Clean-audio leaderboard scores say very little about that world, so I did not rely on them alone.

What I did. I surveyed the 2026 field (Open ASR Leaderboard, arXiv papers, vendor docs — IBM Granite Speech, Cohere Transcribe, NVIDIA Canary/Parakeet, Voxtral, Whisper large-v3, and the commercial APIs), then benchmarked three models myself on one 2-core CPU box: **Whisper base.en** and **Distil-Whisper small.en** (both int8 via faster-whisper) and **Wav2Vec2 base-960h**. Each transcribed the same 36 call-sized LibriSpeech utterances in four conditions: clean, office hum at 10 dB, crowd babble at 10 dB, and babble at 5 dB. I measured word error rate, inference time, memory and setup complexity. The scripts are in the repo; anyone can re-run them.

What I found. Clean WER misleads: on clean audio all three sit within 2.5 points (4.3–5.3%), but at 5 dB babble the spread is 73 points — Wav2Vec2 87.6%, Whisper base.en 25.6%, **Distil-Whisper 14.6%**. Training data explains it: models that saw messy web audio (the Whisper family, 680k hours) held up; the model fine-tuned only on clean read speech fell apart, sometimes inventing more words than the sentence contained. Distil-Whisper also used the least memory (811 MB) and still decoded at 2.5× real time on two CPU cores.

Recommendation. Ship **Distil-Whisper small.en via faster-whisper in int8**. It was the most robust of the three on both speaker tracks, it fits a 1 GB container, it needs one pip install under an MIT license, and the same runtime serves larger distilled models when GPUs arrive. Put a VAD in front, stream partial transcripts, route low-confidence calls to humans, and LoRA fine-tune on 50–200 hours of real calls with noise and telephony-codec augmentation — the biggest cheap win. Wav2Vec2 is out despite being the fastest engine tested; Whisper base.en is the latency-first fallback. Upgrade path on GPU: distil-large-v3 or NVIDIA Parakeet TDT; managed fallback: Deepgram Nova-3. Re-check the leaderboard quarterly — its leadership changed twice in six months.

Caveats. LibriSpeech approximates the acoustics of a call, not its language. 36 utterances is directional, not statistical, though the gaps here (10–70 WER points) are far past noise. CPU speed rankings will shift on GPUs. Before launch I would re-run the same harness on 200+ real call recordings.

Deliverables: technical report, this summary, and the benchmark repo with scripts and README.